

Comparisons of Means Using Exploratory and Confirmatory Approaches

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This article discusses comparisons of means using exploratory and confirmatory approaches. Three methods are discussed: hypothesis testing, model selection based on information criteria, and Bayesian model selection. Throughout the article, an example is used to illustrate and evaluate the two approaches and the three methods. We demonstrate that confirmatory hypothesis testing techniques have more power, that is, have a higher probability of rejecting a false null hypothesis, and confirmatory model selection techniques have a higher probability of choosing the correct or the best hypothesis than their exploratory counterparts. Furthermore, we show that, if more than one hypothesis has to be evaluated, model selection has advantages over hypothesis testing. Another, more elaborate example is used to further illustrate confirmatory model selection. This article concludes with recommendations: When a researcher is able to specify reasonable expectations and hypotheses, confirmatory model selection should be used; otherwise, exploratory model selection should be used.

Keywords: analysis of variance, exploration, confirmation, hypothesis testing, model selection, pairwise comparisons of means

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Researchers are often confronted with the question *Do any of the mean responses differ from the others, and if so, which pairs of means are different from each other?*. For example, Palmer and Gough (2007), examined whether there is a difference in the attribution of importance of defective education as an explanation for criminal behavior between three types of “offenders”: person offenders, property offenders, and non-offenders. The higher the rating, the lower one rates the importance of defective education and the less one considers defective education to be an explanation for criminal behavior. Table 1 presents the descriptive statistics for this example. we use this simple example to describe and illustrate the exploratory and confirmatory methods for comparisons of means. A more elaborate example is discussed at the end of this article.

The model used to answer questions about differences in means is the analysis of variance (ANOVA) model:

$$y_{ij} = \mu_i + \epsilon_{ij}, \quad (1)$$

where y_{ij} is the j th observation ($j = 1, \dots, n_i$) of the dependent variable for group i ($i = 1, \dots, k$), μ_i is the mean of group i , and ϵ_{ij} is the error term. The error terms are independently and normally distributed, with expected value 0 and variance σ^2 , that is, $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$.

There are two different approaches for comparisons of means, namely exploration and confirmation. In exploration, the researcher has no theory or expectation about which pairs or subsets of means are equal. Therefore, all the possible configurations of means are inspected. In the example, where $k = 3$, five hypotheses are under investigation:

$$H_{E0} : \quad \mu_1 = \mu_2 = \mu_3,$$

Table 1: *Number of Observations (n_i), Sample Means ($\bar{y}_i$), and Sample Standard Deviations (sd_i) of the Importance of Defective Education in Explaining Criminal Behavior for Group i*

i offender type	n_i	$\bar{y}_i$	sd_i
1 person	20	11.95	4.42
2 property	20	9.75	3.78
3 non	31	8.77	3.07

$$\begin{aligned}
H_{E1} : & \quad \mu_1 = \mu_2, \mu_3, \\
H_{E2} : & \quad \mu_1, \mu_2 = \mu_3, \\
H_{E3} : & \quad \mu_1 = \mu_3, \mu_2, \\
H_{EA} : & \quad \mu_1, \mu_2, \mu_3,
\end{aligned} \tag{2}$$

where “ $\mu_i = \mu_{i'}$ ”, respectively, “ $\mu_i, \mu_{i'}$ ” denotes that μ_i and $\mu_{i'}$ (for $i, i' = 1, \dots, 3$) are equal, respectively, not restricted. In confirmation, the researcher has one or more theories or expectations with respect to the ordering of the means. In the example, the hypotheses of interest with respect to defective education are:

$$\begin{aligned}
H_{C0} : & \quad \mu_1 = \mu_2 = \mu_3 \\
H_{C1} : & \quad \mu_1 > \mu_2 > \mu_3 \\
H_{CA} : & \quad \mu_1, \mu_2, \mu_3,
\end{aligned} \tag{3}$$

where “ $\mu_i > \mu_{i'}$ ” denotes that μ_i is larger than $\mu_{i'}$ (for $i, i' = 1, \dots, 3$). Hypothesis H_{C1} is based on the expectations of Palmer and Gough (2007) and on previous research. Palmer and Gough expected that the person and property offenders would be less likely to rate defective education as being important for explaining crime (against person or property) than the non-offenders (i.e., $\mu_1 > \mu_3$ and $\mu_2 > \mu_3$). According to previous research, crime against property (e.g., burglary) is likely to be attributed to defective education. Therefore, it is expected that person offenders attribute less importance to defective education than property offenders (i.e., $\mu_1 > \mu_2$). When these expectations are combined, this leads to the hypothesis $H_{C1} : \mu_1 > \mu_2 > \mu_3$. The evaluation of the order-restricted hypothesis H_{C1} requires competing hypotheses. Here, the traditional null (H_{C0}) and alternative (H_{CA}) hypotheses are the competitors.

Within exploration and confirmation, three different types of methods can be distinguished: hypothesis testing, model selection, and Bayesian model selection. Therefore, six different types of techniques for comparisons of means (see Table 2) are introduced and evaluated in this article.

There are several exploratory hypothesis testing techniques. We look at the Shaffer-Welch Fq (SWFq) test, since it is the most powerful test in exploratory hypothesis testing, when controlling α for all comparisons (Ramsey, 2002; Toothaker, 1993, pp. 42–43, 48). Thus, the SWFq test has the highest probability of rejecting a null hypothesis H_0 that is false while controlling the familywise error rate which is defined as $P(\text{at least one false rejection of}$

Table 2: *The Six Techniques for Testing or Comparing Hypotheses (With References)*

Technique	Exploration		Confirmation
Hypothesis testing	Equal n_i :	the Shaffer-Welch Fq test (Toothaker, 1993, pp. 42–43)	The F test (Silvapulle & Sen, 2005, pp. 25–42)
	Unequal n_i :	the Tukey-Kramer test (Toothaker, 1993, pp. 60–61)	
Model selection	Paired-comparison	information criterion (Dayton, 1998, 2003)	Order-restricted information criterion (Anraku, 1999)
Bayesian model selection	Posterior model probabilities		Posterior model probabilities
	(Klugkist, Laudy, & Hoijsink, 2005)		(Klugkist, Laudy, & Hoijsink, 2005)

Note. n_i is the number of observations for group i .

$H_0 : \mu_i = \mu_{i'} \text{ (for all } i, i' = 1, \dots, k) = \alpha$. However, the SWFq test assumes an equal number of observations per group (i.e., $n_i = n$ for $i = 1, \dots, k$). According to Toothaker (1993, pp. 60), the Tukey-Kramer (TK) test is a popular procedure for unequal group observations. The TK test is a modification of the t statistic, a simple technique, and it maintains the α control for all comparisons. Therefore, we look at these two exploratory hypothesis testing techniques, although they might not be as familiar as some other techniques. Note that other techniques, for example, the t test, the Scheffé test, and Fisher’s least significant difference (Toothaker, 1993, pp. 12–13 and 49, 34 and 51, and 41, respectively), are expected to perform similarly. The two most familiar information criteria are probably the Akaike information criterion (AIC) and the Bayesian information criterion (BIC). Another is Rissanen’s information criterion (RIC). When these are used in a classical way, all possible configurations of means are inspected. For $k = 3$, these are summed up in (2). According to Dayton (1998, 2003); Neath and Cavanaugh (2006), the performance of the information criteria can be improved by looking at all the possible configurations of the ordered means. When $k = 3$ and $\bar{y}_1 > \bar{y}_2 > \bar{y}_3$, as in the example (see Table 1), one then does not look at H_{E3} in (2). That is why we look at this exploratory model selection technique, which is called the paired-comparison information criterion (PCIC). Note that the PCIC can be based on any information criterion. We use the AIC, BIC, and RIC, because Dayton (2003) uses these three. In Bayesian model selection (BMS), one can use the posterior model probabilities. Note that BMS is confirmative in essence. It can, however, also be used in an exploratory way. Furthermore, we use the $\bar{F}$ test as confirmatory hypothesis testing technique and the order-restricted information criterion (ORIC) as a confirmatory model selection technique based on an information criterion. As far as we know, these are the only options.

All methods are based on the ANOVA model. The assumptions of the ANOVA model are (a) the dependent variable must be normally distributed for each group, (b) the population variances are equal for each group, and (c) the observations are independent. We assume that these assumptions are met in the examples used in this article. We return to this in the Discussion. In the next section, examples of order-restricted hypotheses encountered in the psychological literature are presented.

Examples of Order-Restricted Hypotheses

In addition to Palmer and Gough (2007), this section presents four other examples of order-restricted hypotheses. At the end of this article another example is given, based upon Gupta, Turban, and Bhawe (2008).

Lievens and Sanchez (2007) investigated the effect of training on the quality of ratings made by consultants. One variable of interest is the signal detection accuracy index, which “refers to the extent to which individuals were accurate in discerning essential from nonessential competencies for a given job” (Lievens & Sanchez, 2007, p. 817). They distinguish three groups of consultants, namely the expert (1), the training (2), and the control (3) group. The authors expected that accuracy of competency ratings would be higher among experts and trained raters than among raters in the control group (i.e., $\mu_1 > \mu_3$ and $\mu_2 > \mu_3$) and furthermore, that it would be highest among raters who already had competency modeling experience (i.e., $\mu_1 > \mu_2$). These expectations can be represented by the hypothesis $H_1 : \mu_1 > \mu_2 > \mu_3$.

Wiener, Holtje, Winter, Cantone, and Gross (2007) examined the purchasing decisions made during a simulated online shopping trip. One of the variables of interest is the likelihood-to-buy. Their design contained two factors. The factor disclosure has two levels: the enhanced (*e*) disclosure condition and the unenhanced (*u*) disclosure condition. In the enhanced disclosure condition, details were given about “credit loan agreements, including interest rates, payment amounts, and repayment time” (Wiener et al., 2007, p. 35); in the unenhanced disclosure condition, no details were given. The factor anticipated emotion has four levels that were manipulated by the researchers: in the pleasant purchase condition (*p*) persons were manipulated to expect a pleasant feeling after a buy and an unpleasant feeling after a nonbuy; in the unpleasant purchase condition (*u*) persons were manipulated to expect an unpleasant feeling after a buy and a pleasant feeling after a nonbuy; in the neutral purchase condition (*n*) persons were manipulated to feel neutral after a buy or a nonbuy; and in the control purchase condition (*c*) the participants simply purchased products without anticipated emotion manipulations. Wiener et al. expected that in both disclosure conditions the pleasant-to-buy group would be more likely to make purchases than the unpleasant-to-buy group (i.e., $\mu_{ep} > \mu_{eu}$ and $\mu_{up} > \mu_{uu}$). Furthermore, they expected that disclosure would be influential only for the control and neutral purchase conditions, where people with “enhanced disclosure in-

formation should buy less than would those without enhanced disclosure” (Wiener et al., 2007, p. 35); $\mu_{ec} > \mu_{uc}$ and $\mu_{en} > \mu_{un}$. The latter implies that no difference is expected in the likelihood-to-buy between the people in the enhanced and unenhanced disclosure groups for both the pleasant-to-buy and unpleasant-to-buy conditions (i.e., $\mu_{ep} = \mu_{up}$ and $\mu_{eu} = \mu_{uu}$). These expectations can be represented by one hypothesis, namely

$$\begin{aligned} H_1 : \quad & \mu_{ec} > \mu_{uc}, \\ & \mu_{en} > \mu_{un}, \\ & \mu_{ep} = \mu_{up} > \mu_{eu} = \mu_{uu} \end{aligned}$$

Hasel and Kassin (2009) examined whether a confession alters the identification decisions of eyewitnesses and their confidence in those decisions. The variable of interest is the confidence rating for their identification. The participants were asked to give a confidence rating after identification (Phase *a*). Then the participants were randomly assigned to four conditions: The participant was told that the identified suspect had confessed (Condition 1), all suspects had denied involvement (Condition 2), the identified suspect had denied involvement (Condition 3), or another person confessed (Condition 4). Hereafter, the participants were asked to give another confidence rating for their identification (Phase *b*). The authors expected that in Condition 1, where the identified suspect confessed, their confidence would increase (i.e., $\mu_{b1} > \mu_{a1}$); that in the other three conditions their confidence would decrease (i.e., $\mu_{b2} < \mu_{a2}$ and $\mu_{b3} < \mu_{a3}$ and $\mu_{b4} < \mu_{a4}$); and that the confidence ratings in Phase *b* would be in decreasing order from Condition 1 to 4 (i.e., $\mu_{b1} < \mu_{b2} < \mu_{b3} < \mu_{b4}$). These expectations can be represented by one hypothesis:

$$\begin{aligned} H_1 : \quad & \mu_{b1} > \mu_{a1}, \\ & \mu_{b2} < \mu_{a2}, \\ & \mu_{b3} < \mu_{a3}, \\ & \mu_{b4} < \mu_{a4}, \\ & \mu_{b1} < \mu_{b2} < \mu_{b3} < \mu_{b4} \end{aligned}$$

Lucas (2003) investigated the difference between female and male leadership. The variable of interest is the influence of the leader, which is measured by the number of times (out of a total of 10) the subject switches to the leader’s answer. Five experimental groups were distinguished: a group with

a randomly selected male leader (Group 1), a group with a randomly selected female leader (Group 2), a group where the male team member who scored highest on a previous task is selected as leader (Group 3), a group where the female team member who scores highest on a previous task is selected as leader (Group 4) and a group in which female leadership is institutionalized and the female team member who scored highest on a previous task is selected as leader (Group 5). The institutionalization was done by showing the participants a film in which it was normal to have female leadership and women did well as leaders. The hypotheses of Lucas (2003) are that male leaders (groups 1 and 3) exert more influence over participants than female leaders in the same leader selection method (Groups 2 and 4, respectively); that is, $\mu_1 > \mu_2$ and $\mu_3 > \mu_4$. Leaders appointed on the basis of their ability (Groups 3 and 4) exerted more influence over participants than leaders of the same sex appointed randomly (Groups 1 and 2, respectively); that is, $\mu_3 > \mu_1$ and $\mu_4 > \mu_2$. Institutionalized female leaders selected on the basis of their ability (Group 5) exerted more influence over participants than “normal” female leaders selected on the basis of their ability (Group 4) and randomly selected female leaders (Group 1); that is, $\mu_3 > \mu_1$ and $\mu_3 > \mu_4$, which can be written (for ease of notation in the resulting H_1) as $\mu_3 > \{\mu_1, \mu_4\}$. Institutionalized female leaders selected on the basis of their ability (Group 5) exerted the same amount of influence over participants as male leaders appointed on the basis of their ability (Group 3); that is, $\mu_5 = \mu_3$. These expectations can be represented by the hypothesis $H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$.

To investigate the hypotheses of interest, software can be helpful. For PCIC only the source code for the programming language GAUSS is made available in an .rtf file (Dayton, 2001). Software for BMS is available, see Klugkist, Laudy, and Hoijtink (2005). The $\bar{F}$ test and the ORIC are not available in any software program. Therefore, these techniques, the SWFq test, and the TK test are implemented in *ComparisonOfMeans.exe*. A zip file with the software *ComparisonOfMeans.exe*, the appropriate text files, and a tutorial can be found at

www.fss.uu.nl/ms/Kuiper and www.fss.uu.nl/ms/informativehypotheses. This zip file also includes data, input, and output files of the examples of Palmer and Gough (2007), Gupta et al. (2008), and Lucas (2003).

In the next sections, we describe the three exploratory and three confirmatory methods and illustrate them using the example concerning the importance of defective education in explaining criminal behavior, along with input and output of *ComparisonOfMeans.exe*. Subsequently, we make two

comparisons: hypothesis testing versus model selection using information criteria and Bayesian model selection, and exploration versus confirmation. The Appendix contains technical notes are provided for the interested reader who wants to learn more about the technical nature of some of the aspects of each technique.

Exploration

Hypothesis Testing Using the SWFq Test

Goal: Find significant pairwise differences.

Procedure: Do an overall F test followed by the Welsch (1977) step-down procedure (Ramsey, 2002; Toothaker, 1993, pp. 42–43). According to Shaffer (1979), the addition of the overall F test increases the power of a step-down procedure. This procedure is elaborated and illustrated below. Note that in Toothaker (1993, pp. 42–43) the technique is called Shaffer-Ryan.

Illustration and Interpretation of the Results: The decision steps of the SWFq test are illustrated using the defective education data of Palmer and Gough (2007). The SWFq test requires equal group sizes. To be able to apply it to the data in Table 1, we generated data with 20 observations for the third group (i.e., the non-offenders) with the same mean and standard deviation as the original 31 cases.

Let n_i , the number of observations in group i , equal n for all i and let N be the total number of observations.

1. **Overall F test:** Do an overall F test. If F is significant, proceed with SWFq; otherwise, stop and fail to reject $H_0 : \mu_1 = \dots = \mu_k$.

In the example, an $F(k-1 = 2, N-k = 57)$ of 3.68 renders a significant p value of 0.03.

2. **Order means (if F is significant) and determine pairwise differences:** Order the means in ascending order. Calculate the pairwise differences (i.e., $\bar{y}_i - \bar{y}_{i'}$ for $i, i' = 1, \dots, k$, where $\bar{y}_i$ is the sample mean of group i) of the ordered means (see Table 3).

3. **Select the largest pairwise difference and determine the stretch size:** Select from all the pairwise differences that have not yet been evaluated the largest pairwise difference $\bar{y}_i - \bar{y}_{i'}$. In case of a tie, select the one with the largest stretch size.

Determine the stretch size s of $\bar{y}_i - \bar{y}_{i'}$. This is the number of means located between $\bar{y}_i$ and $\bar{y}_{i'}$ plus two (for $\bar{y}_i$ and $\bar{y}_{i'}$). One may obtain s by first assigning the numbers 1 to k to the k ordered means such that the numbers indicate the ranking in the ordering and then by calculating

$$s = \text{ranking of } \bar{y}_i - \text{ranking of } \bar{y}_{i'} + 1.$$

In the comparison of the largest and smallest mean, $s = k$. When comparing two adjacent mean, $s = 2$.

In the example, the largest pairwise difference is 3.18 (see Table 3), with a stretch size equal to 3.

4. **Test $H_0 : \mu_i = \mu_{i'}$ (accounting for multiple comparisons):** Reject $H_0 : \mu_i = \mu_{i'}$ when

$$|t_{\bar{y}_i - \bar{y}_{i'}}| = \left| \frac{\bar{y}_i - \bar{y}_{i'}}{\sqrt{MS_W \frac{2}{n}}} \right| \geq t_{crit}^{SWFq} = \frac{q_{df_s, df_W}^{\alpha_s}}{\sqrt{2}}, \quad (4)$$

were MS_W is the within-group mean square of the whole design; that is,

$$MS_W = \frac{\sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2}{N - k},$$

and $q_{df_s, df_W}^{\alpha_s}$ is the critical value of the studentized range distribution with a significance level of

$$\alpha_s = \begin{cases} \alpha, & \text{for } s = k, k-1; \\ 1 - (1 - \alpha)^{s/k}, & \text{for } s \leq k-2, \end{cases}$$

where α is the nominal significance level, often set to .05, the error degrees of freedom $df_W = N - k$, and the "stretch size" degrees of freedom

$$df_s = \begin{cases} k-1, & \text{for } s = k, k-1; \\ s, & \text{for } s \leq k-2. \end{cases}$$

In the example, $|t_{\bar{y}_3 - \bar{y}_1}|$ is 2.65 (see Table 4). The critical value t_{crit}^{SWFq} is 2.00. This implies that $H_0 : \mu_3 = \mu_1$ is rejected. Note that only for $k = 2$ and $k = 3$ are the critical values for all stretch sizes (i.e., when $k = 3$, for $s = 2$ and $s = 3$) the same. For $k > 3$, they are decreasing with the stretch size.

5. **Stopping rule:** When $\bar{y}_i - \bar{y}_{i'}$ is not significant, the subsets of this non-significant set are also nonsignificant. For example, when $k = 3$ and $\bar{y}_1 - \bar{y}_3$ is not significant, the nonsignificant set consists of the Groups 3, 2 and 1, denoted by $\{3, 2, 1\}$. Then, the subsets are $\{3, 2\}$ and $\{2, 1\}$. In other words, when made into a table like Table 3, the pairwise differences below and/or to the left of this nonsignificant pairwise difference are nonsignificant too. Note that these subsets / pairwise differences do not have to be evaluated any more. Return to Step 3 and continue with the largest pairwise difference that has not yet been tested (directly or indirectly). Proceed this way until all pairwise differences are evaluated (directly or indirectly). Table 4 shows that for the example at hand, only $\mu_1 = \mu_3$ is rejected.

Input and Output of ComparisonOfMeans: Note that, in this case, we have three groups with each 20 observations. The data, consisting of two columns, must be given in *Data.txt*. Group number (i) must be given in the first column, and the corresponding data (y_{ij}) must be given in the second column. (More details can be found in the tutorial of ComparisonOfMeans.exe.) When performing the SWFq test, lines 1 to 3 can be ignored, and lines 4 to 9 in *Input.txt* should look as follows:

Table 3: *Ordered Means and Pairwise Differences ($\bar{y}_i - \bar{y}_{i'}$ for $i, i' = 1, 2, 3$) Between the Ordered Means for Defective Education*

Group no. (i)	Ordered means $\bar{y}_i$	$\bar{y}_3 - \bar{y}_i$	$\bar{y}_2 - \bar{y}_i$	$\bar{y}_1 - \bar{y}_i$
3	8.77		0.98	3.18
2	9.75			2.20
1	11.95			

Table 4: *SWFq Test for Defective Education When $n_3 = 20$ (Nominal $\alpha = .05$)*

Pairwise diff. ($\bar{y}_i - \bar{y}_{i'}$)	i & i'	s	$ t_{\bar{y}_i - \bar{y}_{i'}} $	t_{crit}^{SWFq}	Significant
3.18	3 & 1	3	2.65	2.00	yes
2.20	2 & 1	2	1.83	2.00	no
0.98	3 & 2	2	0.82	2.00	no

Note. s = stretch size.

Number of groups
3
Number of observations in the groups
20 20 20
Perform: SWFandTK, PCIC, ExplBMS, Fbar, ORIC, ConfBMS (1 = yes, 0 = no)
1 0 0 0 0 0

All the other lines are not of interest here, but should not be deleted.

In the first lines of the output file, *Output.txt*, there are some remarks regarding the software, followed by the summary of the data. In case of the above mentioned input, the output of the overall F test and the SWFq test is as follows. (Bold numbers change with the data set.)

– Overall F test –

The probability that an F(**2**, **57**) variate is greater than **3.68** is **0.03**, thus reject H0 for all comparisons (at a nominal alpha level of 0.05)

– SWFq –

The following pairs of means are, according to Shaffer-Welch-test, significant different (with **3** groups and **57** degrees of freedom):

Pairwise difference between mean **3** and **1** is significant
(significance level = **0.01**, alpha level = 0.05)

Strengths and Weaknesses: A drawback of the SWFq test is that the SWFq test can render inconsistent results, as can be seen in Table 4. It is logically impossible that $\mu_1 \neq \mu_3$, $\mu_1 = \mu_2$, and $\mu_2 = \mu_3$, because the latter two imply

$\mu_1 = \mu_3$. Another weakness is that when the number of groups (k) and therefore the number of hypotheses increases, the probability of choosing an incorrect hypothesis increases.

Hypothesis Testing Using the TK Test

Goal: Find significant pairwise differences.

Procedure: In the TK test (Tukey, 1953; Kramer, 1956, 1957; Toothaker, 1993, pp. 60–61) a test statistic is calculated for every pairwise difference $\bar{y}_i - \bar{y}_{i'}$ ($i, i' = 1, \dots, k$). Accounting for multiple comparisons, they are evaluated using the studentized range distribution:

$$H_0 : \mu_i = \mu_{i'} \text{ is rejected if } |t_{\bar{y}_i - \bar{y}_{i'}}| = \left| \frac{\bar{y}_i - \bar{y}_{i'}}{\sqrt{MS_W(\frac{1}{n_i} + \frac{1}{n_{i'}})}} \right| \geq t_{crit}^{TK} = \frac{q_{k, df_W}^\alpha}{\sqrt{2}}.$$

Illustration and Interpretation of the Results: In Table 5, the significant and nonsignificant differences for defective education are shown. The same conclusions are obtained as for the SWFq test. Note that the SWFq test is based on a data set with equal ns and the TK test on the original data set.

Input and Output of ComparisonOfMeans: The input for the TK test is the same as for the SWFq test, only in this case the number of observation for Group 3 is 31 and not 20:

Number of observations in the groups
20 20 31

In the example, the output of the TK test looks like this (bold numbers change with the data set):

Table 5: *TK Test for Defective Education (Nominal $\alpha = .05$)*

i & i'	Pairwise diff. ($\bar{y}_i - \bar{y}_{i'}$)	$ t_{\bar{y}_i - \bar{y}_{i'}} $	t_{crit}^{TK}	Significant
1 & 2	2.20	1.89	2.40	no
1 & 3	3.18	3.01	2.40	yes
2 & 3	0.98	0.93	2.40	no

– TK –

The following pairs of means are, according to Tukey-Kramer-test, significant different (with **3** groups and **68** degrees of freedom):

Pairwise difference between mean **1** and **3** is significant
(significance level = **0.01**)

Strengths and Weaknesses: The TK test has the same drawbacks as the SWFq test.

Model Selection Using the PCIC

Goal: Select the best fitting model/hypothesis of a set of hypotheses. The set of hypotheses consists of 2^{k-1} distinct patterns of subsets of ordered means.

Procedure: Calculate $\log L_m$ and PT_m :

- $\log L_m$, that is, the log likelihood for hypothesis H_m , is calculated by:

$$\log L_m(\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}, \hat{\sigma}_m^2 | y) = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log(\hat{\sigma}_m^2) - \frac{1}{2\hat{\sigma}_m^2} \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \hat{\mu}_{mi})^2, \quad (5)$$

where $\hat{\sigma}_m^2 = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \hat{\mu}_{mi})^2$, $N = \sum_i n_i$, and $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$ are the values for which (5) is maximized subject to the restrictions in hypothesis H_m . So the restricted means, that is, $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$, are in accordance with the hypotheses of interest. For example, for hypothesis H_{E0} in (2), it holds that $\hat{\mu}_{01} = \hat{\mu}_{02} = \hat{\mu}_{03} = \bar{\mu}_0$, where $\bar{\mu}_0 = \frac{\sum_{i=1}^3 n_i \bar{y}_i}{\sum_{i=1}^3 n_i}$ is the overall sample mean. For hypothesis H_{E1} in (2), it holds that $\hat{\mu}_{01} = \hat{\mu}_{02} = \frac{n_1 \bar{y}_1 + n_2 \bar{y}_2}{n_1 + n_2}$ and $\hat{\mu}_{03} = \bar{y}_3$. Furthermore, for hypothesis H_{EA} in (2), it holds that $\hat{\mu}_{01} = \bar{y}_1$, $\hat{\mu}_{03} = \bar{y}_3$ and $\hat{\mu}_{02} = \bar{y}_2$.

If the restricted mean values of H_m are equal to the sample mean values, like in hypothesis H_{EA} , H_m has the highest log likelihood value (see Table 6). The larger the difference between the restricted means and the sample means, the lower the value of the log likelihood. For example, the restrictions in hypothesis H_{E2} are not in agreement with

the sample means and the restrictions in H_{E0} even less, therefore, $\log L_2 = -190.13 > \log L_0 = -194.09$ (see Table 6).

- PT_m , that is, the penalty term for hypothesis H_m , is the number of distinct means in H_m (a_m) plus 1 for the residual variance σ^2 , that is, $PT_m = a_m + 1$.

The (log) likelihood represents the fit of the hypothesis with respect to the data at hand. However, the hypothesis with the highest (log) likelihood is not necessarily the best hypothesis. In determining the best hypothesis, the size of the hypothesis also needs to be taken into account. This is represented by the penalty term. As in Ockham's razor, the penalty term can be seen as an implementation of a simple-hypothesis-is-preferred principle. Note that the size of the hypothesis is sometimes referred to as the complexity of the hypothesis. Therefore, the information criteria are determined by the log likelihood and the penalty. As mentioned, we will look at the PCIC (Dayton, 1998, 2003) based on the AIC, the BIC, and the RIC:

$$\begin{aligned} AIC_m &= -2\log L_m + 2(a_m + 1), \\ BIC_m &= -2\log L_m + \log(N)(a_m + 1), \\ RIC_m &= -2\log L_m + \log\left(\frac{N+2}{24}\right)(a_m + 1). \end{aligned}$$

The hypothesis with the smallest value for an information criterion is the preferred hypothesis. This hypothesis is preferred because it best uses the information in the data. In other words, it has the optimal balance between the fit (i.e., $\log L_m$) and the size (i.e., $PT_m = a_m + 1$) of the hypothesis.

Illustration and Interpretation of the Results: There are four hypotheses that can be constructed using the three ordered means of the defective education data: namely H_{E0} , H_{E1} , H_{E2} , and H_{EA} from (2). The AIC, BIC, and RIC values for these hypotheses among others are given in Table 6.

The log likelihood of hypothesis H_{EA} (i.e., $m = A$) is the highest (see Table 6), because for this hypothesis the restricted means (i.e., the $\hat{\mu}_{Ai}$) are equal to the sample means. However, H_{EA} has a complexity factor of four, because it contains three distinct means plus one for the residual variance σ^2 . In this illustration, hypothesis $H_{E2} : \mu_1, \mu_2 = \mu_3$, is the preferred hypothesis (according to all three information criteria).

Table 6: *PCIC of the $2^2 = 4$ Exploratory Hypotheses (H_{Em}) for Defective Education*

Ordered means		8.77	9.75	11.95					
Group nr. (i)		3	2	1					
		Restricted means			PCIC based on				
m	H_{Em}	$\hat{\mu}_{m3}$	$\hat{\mu}_{m2}$	$\hat{\mu}_{m1}$	PT_m	$\log L_m$	AIC_m	BIC_m	RIC_m
0	$\mu_1 = \mu_2 = \mu_3$	9.94	9.94	9.94	2	-196.36	396.71	401.24	394.94
1	$\mu_1 = \mu_2, \mu_3$	8.77	10.85	10.85	3	-193.70	393.41	400.19	390.74
2	$\mu_1, \mu_2 = \mu_3$	9.15	9.15	11.95	3	-192.34	390.68	397.46	388.01
A	μ_1, μ_2, μ_3	8.77	9.75	11.95	4	-191.89	391.79	400.84	388.01

Note. PCIC = paired-comparison information criterion; AIC = Akaike information criterion; BIC = Bayesian information criterion; RIC = Risanen's information criterion. Bolding indicates the lowest value in each column.

Input and Output of ComparisonOfMeans: The input for the PCIC is analogous to the input of the TK test. For the example, the output of the PCIC is given below. In the lines concerning group structure, a number represents a group index. So, "1 2 3" represents μ_1, μ_2, μ_3 , "1 1 1" represents $\mu_1 = \mu_2 = \mu_3$, and "1 2 2" represents $\mu_1, \mu_2 = \mu_3$. (Bold numbers change with the data set.)

– PCIC –

The group-structure and the values of the information criteria for the three best hypotheses, according to the Paired-Comparisons Information Criterion.

AIC = -2 * log likelihood + 2 * penalty
BIC = -2 * log likelihood + log(N) * penalty
RIC = -2 * log likelihood + log((N+2)/24) * penalty

According to the AIC:

Group-structure of the preferred hypothesis: **1 2 2**

Group-structure of the second best hypothesis: **1 2 3**

Group-structure of the third best hypothesis: **1 1 2**

AIC of the preferred hypothesis = $-2 * \mathbf{-192.34} + 2 * \mathbf{3} = \mathbf{390.68}$

AIC of the second best hypothesis = $-2 * \mathbf{-191.89} + 2 * \mathbf{4} = \mathbf{391.79}$

AIC of the third best hypothesis = $-2 * \mathbf{-193.70} + 2 * \mathbf{3} = \mathbf{393.41}$

According to the BIC:

Group-structure of the preferred hypothesis: **1 2 2**

Group-structure of the second best hypothesis: **1 1 2**

Group-structure of the third best hypothesis: **1 2 3**

BIC of the preferred hypothesis = $-2 * \mathbf{-192.34} + \log(N) * \mathbf{3} = \mathbf{397.46}$

BIC of the second best hypothesis = $-2 * \mathbf{-193.70} + \log(N) * \mathbf{3} = \mathbf{393.41}$

BIC of the third best hypothesis = $-2 * \mathbf{-191.89} + \log(N) * \mathbf{4} = \mathbf{391.79}$

According to the RIC:

Group-structure of the preferred hypothesis: **1 2 2**

Group-structure of the second best hypothesis: **1 2 3**

Group-structure of the third best hypothesis: **1 1 2**

RIC of the preferred hypothesis =
 $-2 * \mathbf{-192.34} + \log((N+2)/24) * \mathbf{3} = \mathbf{388.01}$

RIC of the second best hypothesis =
 $-2 * \mathbf{-191.89} + \log((N+2)/24) * \mathbf{4} = \mathbf{388.24}$

RIC of the third best hypothesis =
 $-2 * \mathbf{-193.70} + \log((N+2)/24) * \mathbf{3} = \mathbf{390.74}$

Note that the results of the PCIC is consistent with the results of the TK test.

Strengths and Weaknesses: The strength of the PCIC is that it, like all model selection techniques, always gives consistent results. A weakness of the PCIC is that when the number of groups (k) and therefore the number of hypotheses (2^{k-1}) increases, the probability of selecting the best hypothesis decreases. Because in model selection there is no null hypothesis, we do not call this

probability power but the probability of choosing the best hypothesis.

Model Selection Using Exploratory BMS

Goal: Select the best fitting model/hypothesis of a set of hypotheses. In exploratory BMS, like when using the PCIC, the set of hypotheses consists of 2^{k-1} distinct patterns of subsets based on means ordered from smallest to largest.

Procedure: Calculate the marginal likelihood of a hypothesis. In order to do this, the likelihood with respect to the parameters $\mu_1, \dots, \mu_k, \sigma^2$, and the prior distribution of these parameters are needed. The log likelihood is given in (5).

Prior Distribution. The prior distribution reflects prior knowledge with respect to the means $(\mu_1, \dots, \mu_k)$ and the residual variance (σ^2) . So, the (order) restrictions of H_m are taken into account in the prior (see the technical note “The Prior” in the Appendix for an elaboration). Klugkist, Laudy, and Hoi-jtink (2005) used the same and mutually independent prior distribution for each mean. They and others (Chen & Sungduk, 2008; Johnson, 2005; Rossell, Baladandayuthapani, & Johnson, 2008) showed that such a prior does not favor any of the hypotheses under investigation. Furthermore, they showed that this prior has good properties when the goal is to select the best of a set of order-restricted hypotheses.

The prior knowledge of each mean is represented by a normal distribution, with a data-based mean β_0 and data-based variance τ_0^2 . The prior distribution of σ^2 is a scaled inverse chi-squared distribution, with hyperparameters ν_0 and κ_0^2 , where ν_0 is the degrees of freedom parameter and κ_0^2 is the scale parameter. (For an elaboration, see the technical note “Data-based Hyperparameters” in the Appendix.) The prior is chosen such that it has a minimal impact on the results. This is done by choosing a prior which is not only vague (such that it has low impact), but also compatible with the data (such that it is not too vague; Klugkist, Laudy, & Hoi-jtink, 2005).

The mean β_0 and the variance τ_0^2 depend not only on the data but also on a user-specified term that reflects the vagueness of the prior (PV). Three degrees of vagueness are used here: PV equal to 1, 2, and 3, where $PV = 3$ renders the vaguest prior.

Marginal Likelihood. The marginal likelihood for hypothesis H_m is a measure of the degree of support for hypothesis H_m provided by the data (Klugkist, Laudy, & Hoijsink, 2005). It is equal to the integral of the likelihood over the prior distribution for the hypothesis at hand, H_m (see the technical note “The Marginal Likelihood” in the Appendix for an elaboration). For an elaboration of the interpretation and calculation of the marginal likelihood, we refer the reader to Klugkist (2008) and Klugkist, Laudy, and Hoijsink (2005).

The marginal likelihood quantifies the support in the data for the hypothesis at hand (H_m) accounting for the fit and complexity/size of H_m . In this way, the marginal likelihood resembles information criteria, like the PCIC based on the AIC, which can be written as $-2 \log \text{likelihood} + 2 \text{penalty}$. That is, -2 times the log of the marginal likelihood can be written as $-2 \log \text{likelihood} + 2 \text{penalty}$ (see the technical note “The Marginal Likelihood” in the Appendix), where the penalty equals -1 times the log of the ratio of the prior density and the posterior density. Note that the marginal likelihood is higher for a hypothesis with a better fit and/or a lower complexity.

To interpret several marginal likelihoods at once, it can be helpful to transform them into posterior model probabilities (PMPs). A PMP is the probability that, given the data, the corresponding hypothesis is the best of the set of hypotheses (assuming a priori that all the hypotheses have equal probabilities of being the best; the interested reader is referred to the technical note “Posterior Model Probability” in the Appendix). The hypothesis with the highest PMP is the preferred hypothesis, that is, the best hypothesis of the set of hypotheses.

Illustration and Interpretation of the Results: In exploratory BMS, the same hypotheses are compared as in PCIC. In Table 7, the PMPs (for each hypothesis and each prior vagueness) are shown for the defective education data. The preferred hypothesis (for each prior vagueness) is hypothesis $H_{E2} : \mu_1, \mu_2 = \mu_3$.

Note that when PV increases, the support for hypotheses with equality constraints (“=”), that is, H_{E0} , H_{E1} , and H_{E2} , relative to the support for the unconstrained model (i.e., H_{EA}) increases. However, the differences in PMP values are not that large for the different PV values. This implies that the results are robust with respect to the specification of the prior distribution. The interested reader is referred to Hoijsink et al. (2008), Klugkist, Laudy, and Hoijsink (2005), and Klugkist, Kato, and Hoijsink (2005) for further

Table 7: *Posterior Model Probabilities (PMP) of the $2^2 = 4$ Exploratory Hypotheses (H_{Em}) for Defective Education for Three Types of Prior Vagueness (PV)*

m	H_{Em}	PMP		
		PV = 1	PV = 2	PV = 3
0	$\mu_1 = \mu_2 = \mu_3$	.04	.06	.06
1	$\mu_1 = \mu_2, \mu_3$	.13	.16	.17
2	$\mu_1, \mu_2 = \mu_3$	.58	.55	.62
A	μ_1, μ_2, μ_3	.25	.23	.16

Note. Bolding indicates the highest value in each column.

elaboration.

Input and Output of ComparisonOfMeans: The input for exploratory BMS is analogous to the input for the TK test. However, in case of BMS, two additional specifications are needed: the desired δ and the prior vagueness PV . Specify $\delta = 0$ for an “exact equality” (i.e., $\mu_1 = \dots = \mu_k$, that is, $|\mu_i - \mu_{i'}| = \delta = 0$ for all $i, i' = 1, \dots, k$) and any positive number (i.e., $\delta > 0$) for an “about equality” (i.e., $\mu_1 \approx \dots \approx \mu_k$, that is, $|\mu_i - \mu_{i'}| < \delta$ for all $i, i' = 1, \dots, k$). In the latter case, one must specify δ in a reasonable way, for example, by looking at previous studies and/or asking experts in the field. The default recommendation for PV is $PV = 2$, but any positive number may be specified. When setting $\delta = 0$ and $PV = 3$, the first lines of *Input.txt* should look as follows:

When BMS is performed, an interval for equality relations (delta) is needed and a parameter for prior vagueness (pv).
When BMS is not performed, nothing needs to be filled in (but do not delete these lines).
0.0 3.0

In the example, the output of exploratory BMS looks like the following. (Bold numbers change with the data set.)

– exploratory BMS –

The preferred hypothesis, according to exploratory Bayesian model selection, has the following group-structure:

1 2 2

The resulting Bayes factor (BF) value (of this hypothesis versus the unconstrained hypothesis) and the posterior model probabilities (PMP) of this hypothesis with respect to the whole set of hypotheses:

PMP

0.62

As in PCIC, “1 2 2” represents $\mu_1, \mu_2 = \mu_3$.

Strengths and Weaknesses: The strength of BMS is that it, like all model selection techniques, always gives consistent results. A weakness of BMS is that it is a time consuming technique. In addition, if a hypothesis contains an equality constraint (i.e., “=”), BMS can be sensitive to the choice of the prior. However, as shown in Table 7, for these choices of PV this sensitivity usually does not lead to a different evaluation of the hypotheses under investigation.

Confirmation

Hypothesis Testing Using the $\bar{F}$ Statistic

In standard statistical testing, the hypothesis *all means are equal* (i.e., $H_0 : \mu_1 = \dots = \mu_k$) is tested against the alternative *not all means are equal* (i.e., $H_A : \mu_1, \dots, \mu_k$). This is usually tested with an F test using a one-way ANOVA. However, researchers often want to test a certain order restriction because of a theory or expectation with respect to the order of the means in the experiment. For example, it is expected that non-offenders are more likely to rate defective education as being important for explaining crime than the other two offenders, and property offenders are more likely to rate it as being important than person offenders (see H_{C1} in (3)). Such an order-restricted hypothesis can be tested with the $\bar{F}$ test.

Goal: Evaluate the null hypothesis. When testing an *ordered alternative* (i.e., test *all means are equal* against an order-restricted hypothesis), the traditional null hypothesis is evaluated. When testing an *ordered null* (i.e.,

test an order-restricted hypothesis against *all parameters are free*), the order-restricted hypothesis is evaluated. In summary, in the ordered alternative, $H_0 : \mu_1 = \dots = \mu_k$ is tested against $H_m : \mu_i - \mu_{i'} \geq 0$ (for some $i, i' = 1, \dots, k$) and, in the ordered null, $H_m : \mu_i - \mu_{i'} \geq 0$ (for some $i, i' = 1, \dots, k$) is tested against $H_A : \mu_1, \dots, \mu_k$.

Procedure: Calculate the value of the $\bar{F}$ statistic (Silvapulle & Sen, 2005, pp. 25–42). Like the classical F test, the $\bar{F}$ test is based on the residual sum of squares (RSS) for the tested null distribution (i.e., the classical null or an order-restricted hypothesis) and the tested alternative (i.e., an order-restricted hypothesis or the classical alternative, respectively). For the classical null (H_0) and the classical alternative (H_A), the RSS are determined with respect to the overall mean ($\bar{y}$) and sample means ($\bar{y}_i$), respectively. Note that these are the values for which the RSS is minimized given that the values are in accordance with the hypothesis at hand (i.e., H_0 and H_A , respectively). The analogue is done for an (order-restricted) hypothesis, H_m . The values that minimize the RSS given that these are in accordance with H_m are called the restricted means. (For an elaboration see the technical note “The $\bar{F}$ Statistic” in the Appendix.) So, the order restrictions are taken into account in one of the RSS (depending on the type of $\bar{F}$ test).

As with classical hypothesis testing, p values must be determined. Because of the order-restrictions, this is done via simulation (Silvapulle & Sen, 2005, pp. 32–33 and 40; for an elaboration see the technical note “Calculation of the p Value of the $\bar{F}$ Statistic” in the Appendix).

Illustration and Interpretation of the Results: Table 8 shows that $H_{C0} : \mu_1 = \mu_2 = \mu_3$ is rejected (for $\alpha = 0.05$) when it is tested against both $H_{CA} : \mu_1, \mu_2, \mu_3$ and $H_{C1} : \mu_1 > \mu_2 > \mu_3$ and that H_{C1} is not rejected when it is tested against H_{CA} . So, H_{CA} is preferred over H_{C0} and H_{C1} is preferred over both H_{C0} and H_{CA} . Therefore, H_{C1} is the preferred hypotheses.

Input and Output of ComparisonOfMeans: For all three confirmatory techniques (i.e., the $\bar{F}$ test, ORIC, and confirmatory BMS), all hypotheses of interest must be given explicitly. This must be done in *Input.txt* from line 10 on. In the example, the hypotheses of interest are the set of hypotheses specified in (3).

In order to write down the hypotheses of interest in the input file, some notation must be introduced. The way the hypotheses are written down

Table 8: $\bar{F}$ Test of the Specified Hypotheses for Defective Education

Hypotheses tested			$\bar{F}$	p
H_{C0}	against	H_{CA}	9.11	.01
H_{C0}	against	H_{C1}	9.11	.00
H_{C1}	against	H_{CA}	0.00	1.00

in the input file has to do with the hypothesized group numbers and the inequality constraints “>” and “<”. For ease, the first mean is said to be in the first group. When looking at $H_{C0} : \mu_1 = \mu_2 = \mu_3$, all means are hypothesized to be in the same group. Therefore, H_{C0} is written down as “1 1 1”. When looking at $H_{CA} : \mu_1, \mu_2, \mu_3$, all means are hypothesized to be in another group. Because there are no restrictions on the means, H_{CA} is written down as “1 2 3”. When looking at $H_{C1} : \mu_1 > \mu_2 > \mu_3$, all means are also hypothesized to be in another group, but now there are restrictions on the means. The sign “>” is represented by a “-3”. Therefore, $H_{C1} : \mu_1 > \mu_2 > \mu_3$ is represented by “1 -3 -3”, meaning that the first mean is bigger than the second, which is bigger than the third one. In the example, the ordering of the group numbers for all three hypotheses is “1 2 3” and all three hypotheses are written down in one line/restriction. (More details on this can be found in the tutorial of *ComparisonOfMeans.exe*.)

When performing the $\bar{F}$ test, the first half of *Input.txt* is analogous to the input of the TK test, and the second half, from line 10 on, should look as follows:

In case of Fbar test and/or ORIC and/or ConfBMS: Number of hypotheses to be compared 3 Number of restrictions per hypothesis 1 1 1 Ordering of means in restriction 1 2 3 1 2 3

1	2	3
(Order) Restrictions		
1	1	1
1	-3	-3
0	0	0

The output of the $\bar{F}$ test in the example looks like this (bold numbers change with the data set and/or set of hypotheses):

– Fbar test –

Results of the Fbar test for the null hypothesis **1** and the unconstrained hypothesis **3**

Hypotheses numbers	Fbar value	p-value
1 versus 3	9.11	0.01

Results of the "ordered alternative" Fbar test

Ordered-hypothesis number	Fbar value	p-value
H0 versus 2	9.11	0.00

Results of the "ordered null" Fbar test

Ordered-hypothesis number	Fbar value	p-value
2 versus Ha	0.00	1.00

Strengths and Weaknesses: A disadvantage of the $\bar{F}$ test is that it can only test one order-restricted hypothesis at a time. In this case, the $\bar{F}$ test results in three p values (see Table 8), which can be evaluated straightforwardly and will give a consistent result. When a researcher wants to test more than one order-restricted hypothesis, say H_1 and H_2 , problems arise. Then the $\bar{F}$ tests results in five p values: one resulting from the classical F test, two from the ordered alternative for both H_1 and H_2 , and two from the ordered null for both H_1 and H_2 . No straightforward, nonarbitrary rules exist for the evaluation of these five p values. Furthermore, these p values cannot be used for a direct comparison of the two order-restricted hypotheses H_1 and H_2 . Therefore, the $\bar{F}$ test is most useful when one order-restricted hypothesis has to be evaluated.

Model Selection Using the ORIC

Goal: Select the best fitting model/hypothesis of a set of hypotheses. The ORIC inspects a limited set of well defined hypotheses constructed using one or more restrictions of the form $\mu_i - \mu_{i'} \geq 0$ for $i, i' = 1, \dots, k$.

Procedure: Calculate the ORIC (Anraku, 1999) for hypothesis H_m by

$$ORIC_m = -2 \log L_m + 2 PT_m, \quad (6)$$

where

- $\log L_m$ is calculated by (5).

Note that in the ORIC the order restrictions are taken into account in the (order-restricted) likelihood. Namely, the (order-restricted) likelihood is based on means that maximize the likelihood (like the maximum likelihood estimates), only these estimates are in accordance with the restrictions in H_m . We will refer to these means as the restricted means. Note that the restricted means $(\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk})$ are the same as the restricted means in the $\bar{F}$ test (see the technical note “The Restricted Means” in the Appendix).

- PT_m is calculated by:

$$PT_m = 1 + \sum_{l=1}^{a_m} LP_{ml} \cdot l, \quad (7)$$

where LP_{ml} is the level probability for hypothesis H_m , that is, the a priori probability that there are l distinct mean values among $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$ (which are in agreement with H_m), and a_m equals the number of distinct μ_i s in the restrictions of H_m . For example, in $H_1 : \mu_1 > \mu_2 > \mu_3$, $a_1 = 3$ and, in $H_2 : \mu_1 > \mu_2 = \mu_3$, $a_2 = 2$. The computation of the level probabilities can be done via simulation (Silvapulle & Sen, 2005, pp. 78–81). For a description of this simulation, see the technical note “Calculation of the Level Probabilities” in the Appendix.

The order restrictions are not only taken into account in the (order-restricted) likelihood, but also in the penalty term. The penalty term can be seen as the expected number of distinct mean values plus 1 (because of the unknown variance term). When there are no order

Table 9: *ORIC of the Three Specified Hypotheses (H_{Cm}) for Defective Education*

m	H_{Cm}	Restricted means			PT_m	$\log L_m$	$ORIC_m$
		$\hat{\mu}_{m1}$	$\hat{\mu}_{m2}$	$\hat{\mu}_{m3}$			
0	$\mu_1 = \mu_2 = \mu_3$	9.94	9.94	9.94	2.00	-196.36	396.71
1	$\mu_1 > \mu_2 > \mu_3$	11.95	9.75	8.77	2.82	-191.89	389.42
A	μ_1, μ_2, μ_3	11.95	9.75	8.77	4.00	-191.89	391.79

Note. ORIC = order-restricted information criterion. Bolding indicates the lowest value.

restrictions, this penalty reduces to the penalty of the PCIC, that is, $PT_m = a_m + 1$, and ORIC reduces to the AIC.

The hypothesis with the smallest value for the ORIC is the preferred hypothesis (as in the PCIC).

Illustration and Interpretation of the Results of the ORIC: In Table 9, the restricted means ($\hat{\mu}_{mi}$), the log likelihood values ($\log L_m$), the penalty terms (PT_m), and the ORIC values are given for the three hypotheses for defective education.

Since the sample means are in accordance with the restrictions in both H_{C1} and H_{CA} , the restricted means (i.e., the $\hat{\mu}_{mi}$'s for $m = 1$ and $m = A$) are equal to the sample means. Therefore, hypotheses H_{C1} and H_{CA} have the highest log likelihood. However, H_{C1} is less complex than H_{CA} (i.e., $PT_1 < PT_A$), because the means in H_{CA} are not restricted. When looking at the optimal combination of fit (i.e., $\log L_m$) and size/complexity (i.e., PT_m) of the hypotheses, H_{C1} is the preferred hypothesis.

Input and Output of ComparisonOfMeans: The input for the ORIC is analogous to the input for the $\bar{F}$ test. In the example, the output of the ORIC looks like this (bold numbers change with the data set and/or set of hypotheses):

– ORIC –

The value of the Order-Restricted Information Criterion (ORIC) =
-2 * log likelihood + 2 * penalty:

for Hypothesis 1, ORIC = -2 * **-196.36** + 2 * **2.00** = **396.71**

for Hypothesis 2, ORIC = -2 * **-191.89** + 2 * **2.82** = **389.42**

for Hypothesis 3, ORIC = -2 * **-191.89** + 2 * **4.00** = **391.79**

The preferred hypothesis, according to the Order-Restricted
Information Criterion, of the hypotheses to be compared is
hypothesis number **2**,

with the following ordering(s) of means:

1 2 3

and corresponding restriction(s):

1 -3 -3

Strengths and Weaknesses of the ORIC: As all model selection techniques, the ORIC always gives consistent results.

Most researchers are able to specify reasonable hypotheses, since they are expert in their research field. However, it is possible that the set of specified hypotheses does not contain a reasonable or good hypothesis. In that case, a model selection technique, like ORIC, will choose the best hypothesis of a set of weak hypotheses. To ensure that a weak hypothesis is not chosen, one can include the unconstrained hypothesis $H_A : \mu_1, \dots, \mu_k$ in the set of hypotheses. H_A will always be preferred over a weak order-restricted hypothesis.

Model Selection Using Confirmatory BMS

Goal: Select the best fitting model/hypothesis of a set of hypotheses. Confirmatory BMS, like the ORIC, examines a limited set of well defined hypotheses constructed using one or more restrictions of the form $\mu_i - \mu_{i'} \geq 0$ for $i, i' = 1, \dots, k$.

Procedure: Calculate the marginal likelihood of a hypothesis as is done in

exploratory BMS. Note that, in BMS, the order restrictions are taken into account via the admissible space of the prior for hypothesis H_m (more details can be found in the technical note “The Prior” in the Appendix).

Illustration and Interpretation of the Results: In Table 10, the PMPs (for each prior vagueness) are given for the three hypotheses for defective education. The preferred hypothesis (for each prior vagueness) is $H_{C1} : \mu_1 > \mu_2 > \mu_3$. Note again the robustness of the inferences with respect to the vagueness of the prior distribution.

Input and Output of ComparisonOfMeans: The input for confirmatory BMS is analogous to the input for the $\bar{F}$ test. Note that, in this case, as when using exploratory BMS, δ and PV must be specified. In the example, with $\delta = 0$ and $PV = 3$, the output of confirmatory BMS looks like this (bold numbers change with the data set and/or set of hypotheses):

Table 10: *Posterior Model Probabilities (PMP) of the Three Specified Hypotheses (H_{Cm}) for Defective Education for Three Types of Prior Vagueness (PV)*

m	H_{Cm}	<i>PMP</i>		
		$PV = 1$	$PV = 2$	$PV = 3$
0	$H_{C0} : \mu_1 = \mu_2 = \mu_3$	.03	.03	.06
1	$H_{C1} : \mu_1 > \mu_2 > \mu_3$	.80	.80	.78
A	$H_{CA} : \mu_1, \mu_2, \mu_3$	.17	.17	.17

Note. Bolding indicates the highest value in each column.

– confirmatory BMS –

The resulting posterior model probabilities (PMP) of the order-restricted hypotheses with respect to the whole set of hypotheses:

	PMP
Hypothesis 1	0.07
Hypothesis 2	0.77
Hypothesis 3	0.16

The preferred hypothesis, according to confirmatory Bayesian model selection, of the hypotheses to be compared is hypothesis number **2**, with the following ordering(s) of means:

1 2 3

and corresponding restriction(s):

1 -3 -3

Strengths and Weaknesses: Confirmatory BMS gains and suffers from the same things as exploratory BMS. Also here, the illustration (see Table 10) shows that for reasonable choices of PV , the prior sensitivity does usually not lead to a different evaluation of the hypotheses. Furthermore, if a hypothesis contains only inequality constraints (i.e., “<” and/or “>”), the relative support of this hypothesis with respect to the unconstrained hypothesis shows that BMS is not sensitive to the choice of the prior.

As is the case when using the ORIC, the unconstrained hypothesis should be included in the set of hypotheses to protect against choosing a weak order-restricted hypothesis.

Conclusions with Respect to Defective Education

From the SWFq test (in case of equal group sizes) and the TK test, it follows that only the pairwise difference between Groups 3 and 1 is significant. Palmer and Gough (2007) concluded the same when performing an ANOVA F test and a Scheffé post hoc test. As mentioned before, this is a logically impossible result, because $\mu_1 = \mu_2$ and $\mu_2 = \mu_3$ would imply $\mu_1 = \mu_3$. When using the PCIC and exploratory BMS, it can be concluded that hypothesis $H_{E2} : \mu_1, \mu_2 = \mu_3$ is the preferred hypothesis (according to all three criteria and for each prior vagueness, respectively). When looking at the sample

Table 11: *TK Test for Defective Education when $n_3 = 20$ (Nominal Alpha = 0.05)*

i & i'	Pairwise diff. $(\bar{y}_i - \bar{y}_{i'})$	$ t_{\bar{y}_i - \bar{y}_{i'}} $	t_{crit}^{TK}	Sign.
3 & 1	3.18	2.65	2.41	yes
2 & 1	2.20	1.83	2.41	no
1 & 2	0.98	0.82	2.41	no

means, more can be said about the ordering of the means. However, when using an exploratory method, the hypotheses of interest, $\mu_1 > \mu_2 > \mu_3$, is not evaluated. From the three $\bar{F}$ tests (i.e., the classical F test, the ordered alternative, and the ordered null), we concluded that $H_{C1} : \mu_1 > \mu_2 > \mu_3$ is the preferred hypotheses. The same is concluded when using the ORIC or doing confirmatory BMS (for each prior vagueness).

Comparison of the Two Approaches and the Three Methods

The TK Test is Less Powerful than the SWFq Test

In case of equal group sizes, the TK test is less powerful than the SWFq test. Thus, the SWFq test has a higher probability of rejecting a false null hypothesis than the TK test has. This can be seen from the critical values for both tests (Toothaker, 1993, pp. 42–43, 48), because both tests are based on the same calculated value $|t_{\bar{y}_i - \bar{y}_{i'}}|$. In the example, in case of equal group sizes in defective education (i.e., $n_1 = n_2 = n_3 = 20$), the SWFq test has a critical value of 2.00 (see Table 4) which is lower than 2.41 (see Table 11), the critical value of the TK test. So, when $2.41 > |t_{\bar{y}_i - \bar{y}_{i'}}| > 2.00$ (for $i, i' = 1, 2, 3$), the SWFq test does reject the (false) null hypothesis and the TK test does not.

The $\bar{F}$ Test is More Powerful than the SWFq Test or the F Test

The advantage of the $\bar{F}$ test is that it has more power than the SWFq test, since it tests fewer hypotheses and it can test order-restricted hypotheses. The increase in power due to testing order-restricted hypotheses can be il-

illustrated by comparing the ordered alternative $\bar{F}$ test with the classical F test.

For example, look at a generated data set with $k = 3$ groups, effect size ES , and n observations per group. When $ES = 0$, the sample means are the same and, when $ES = 0.1, 0.2, 0.3, 0.4$, or 0.5 , the sample means are decreasing, that is, $\bar{y}_1 > \bar{y}_2 > \bar{y}_3$. Note that $ES = \frac{1}{\hat{\sigma}} \sqrt{\frac{1}{3} \sum_{i=1}^3 (\bar{y}_i - \bar{y})^2}$ (Cohen, 1992), where $\bar{y} = \frac{1}{3} \sum_{i=1}^3 \bar{y}_i$ and $\hat{\sigma}$ is set to 1. An effect size of $ES = 0.1$ is called small, $ES = 0.25$ medium, and $ES = 0.4$ large. For $ES = 0$, a sample is constructed that is perfect in agreement with $H_0 : \mu_1 = \mu_2 = \mu_3$, and for $ES > 0$, one that is perfect in agreement with $H_1 : \mu_1 > \mu_2 > \mu_3$. In the ordered alternative $\bar{F}$ test, $H_0 : \mu_1 = \mu_2 = \mu_3$ is tested against $H_1 : \mu_1 > \mu_2 > \mu_3$. Note that, in the F test, $H_0 : \mu_1 = \mu_2 = \mu_3$ is tested against $H_A : \mu_1, \mu_2, \mu_3$.

In Table 12, the p values of both tests are given for each generated data set. For $ES > 0$, the p values of the ordered alternative $\bar{F}$ are always lower than the p values of the F test. Thus, (the false) H_0 is rejected for an lower effect size when tested against H_1 than when tested against H_A . Therefore, the ordered alternative $\bar{F}$ test has more power than the F test (when the order-restricted hypothesis is true). Hence, testing order-restricted hypotheses increases power.

What if another order-restricted hypothesis, say H_2 , is true? In that case, if H_0 is tested against H_1 , H_0 may or may not be rejected. However, if H_1 is tested against H_A , H_1 will be rejected (if the sample is large enough).

The ORIC is More “Powerful” than the PCIC

The ORIC has an higher probability of choosing the best hypothesis than the PCIC, because it evaluates fewer hypotheses and it can evaluate order-restricted hypotheses. We illustrate the latter by comparing the ORIC with the AIC, using the same data sets used to illustrate the gain in power when the $\bar{F}$ test is used instead of the F test.

The three hypotheses to be evaluated are:

$$\begin{aligned} H_0 : & \quad \mu_1 = \mu_2 = \mu_3, \\ H_1 : & \quad \mu_1 > \mu_2 > \mu_3, \\ H_A : & \quad \mu_1, \mu_2, \mu_3. \end{aligned}$$

In Figure 1, the ORIC values are given for these three hypotheses (for $n = 10$

Table 12: p values of the Ordered Alternative $\bar{F}$ Test ($p_{\bar{F}}$) and the F Test (p_F) for One Generated Data Set With $k = 3$ Groups, Effect Size ES , and n Observations per Group

p value	n			
	10	20	50	100
$ES = 0$				
p_F	1.00	1.00	1.00	1.00
$p_{\bar{F}}$	1.00	1.00	1.00	1.00
$ES = 0.1$				
p_F	0.91	0.82	0.61	0.37
$p_{\bar{F}}$	0.48	0.40	0.26	0.14
$ES = 0.2$				
p_F	0.6742	0.4543	0.1390	0.0193
$p_{\bar{F}}$	0.3022	0.1802	0.0470	0.0059
$ES = 0.3$				
p_F	0.4184	0.1746	0.0127	0.0002
$p_{\bar{F}}$	0.1648	0.0599	0.0039	0.0000
$ES = 0.4$				
p_F	0.220454	0.048182	0.000501	0.000006
$p_{\bar{F}}$	0.080770	0.015600	0.000110	0.000000
$ES = 0.5$				
p_F	0.100899	0.009983	0.000011	0.000001
$p_{\bar{F}}$	0.034460	0.002880	0.000000	0.000000

and $n = 100$). As mentioned before, the ORIC reduces to the AIC when there are no order restrictions, that is, for hypotheses H_0 and H_A . The ORIC values for H_0 increase with ES , because the differences between the sample means $\bar{y}_i$ and the restricted means $\hat{\mu}_{0i} = \bar{\mu}_0$ (for all $i = 1, \dots, k$) increase with increasing effect size. This leads to a decrease in the log likelihood (see (5)) and, consequently, an increase in ORIC value (see (6)). The ORIC values for H_1 and H_A do not depend on effect size, because the sample means are in accordance with H_1 and, logically, H_A . Thus, the difference between $\bar{y}_i$ and $\hat{\mu}_{mi}$ (for $i = 1, \dots, k$) is zero for both H_1 and H_A for each effect size. This implies that the likelihood values for H_1 and H_A are equal; therefore, the difference in ORIC values equals two times the difference in the penalty terms, that is, $2(PT_2 - PT_1) = 2(4 - 2\frac{5}{6}) = 2\frac{1}{3}$. So, when the sample means are in accordance with H_1 , as in the example, H_1 is always preferred over H_A . As can be seen in Figure 1, compared to H_A , H_1 will be preferred over H_0 for smaller effect sizes. This implies that the probability of choosing the correct/best hypothesis is higher if H_0 is compared to H_1 than if H_0 is compared to H_A (when H_1 is true).

In this example, the ideal case is represented, namely H_1 is true and H_1 is included in the set of hypotheses. What if another order-restricted hypothesis, say H_2 , is true? In that case, H_A will be preferred over H_1 if the sample size is large enough to distinguish H_1 from H_2 . Therefore, the unconstrained model H_A should always be included in the set of hypotheses. So, if the unconstrained hypothesis is included in the set, model selection techniques select the correct hypothesis or a similar one (i.e., a hypothesis that resembles the true hypothesis, that is, only differs in a few constraints) or otherwise the unconstrained hypothesis.

Exploratory BMS is more “Powerful” than Confirmatory BMS

Confirmatory BMS has the same advantages over exploratory BMS as the ORIC has over the AIC.

Conclusion of Comparisons

In confirmation there will be more power (i.e., a higher probability of rejecting a false null hypothesis) or a higher probability of choosing the best hypothesis

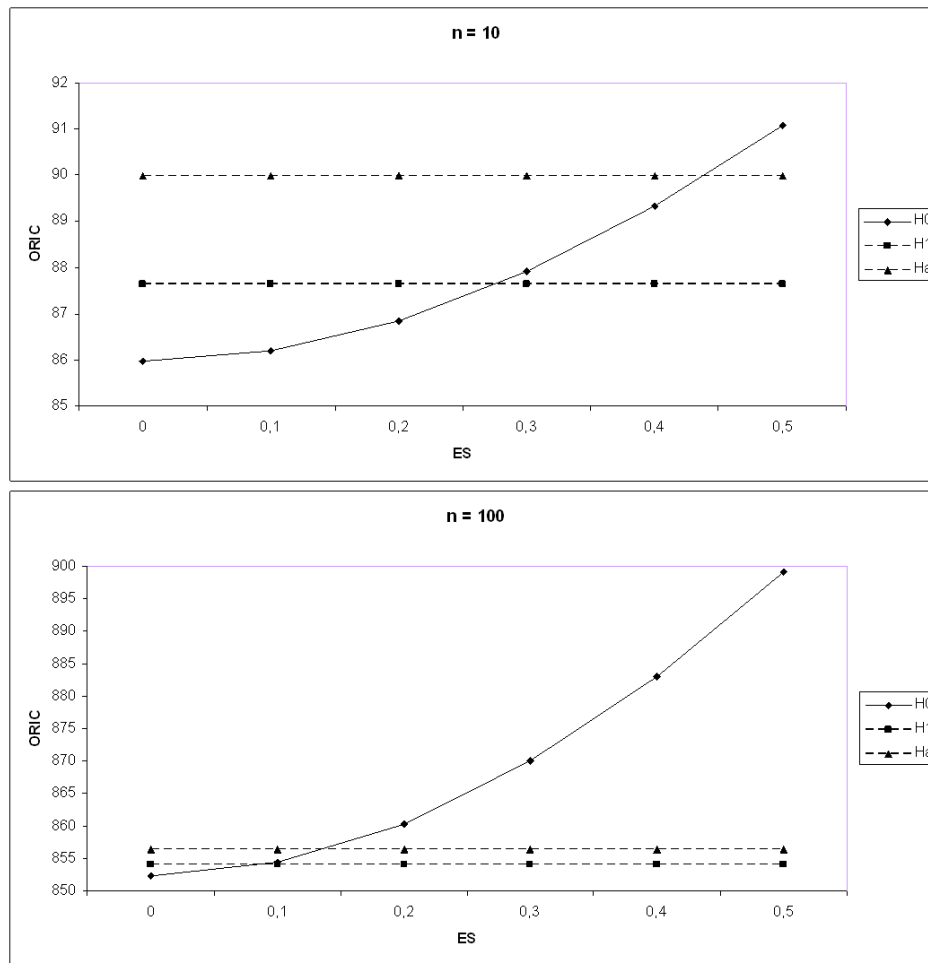


Figure 1: Order-restricted information criterion (ORIC) values for one generated data set with $k = 3$ groups, effect size ES , and n observations per group.

than in exploration. This is due to the smaller number of hypotheses that have to be evaluated and to the inclusion of expectations, in the form of order restrictions, in the hypotheses of interest (see Table 12 and Figure 1). But, the most important advantage is that the expectations of researchers (see the examples given in the beginning of this article) can be evaluated using a confirmatory approach. The weakness of confirmation is that a researcher must be able to specify his expectations. Note that the inclusion of the unconstrained hypothesis $H_A : \mu_1, \dots, \mu_k$ is a safeguard against choosing a hypothesis that is weakly supported by the data.

Model selection, in contrast to hypothesis testing, always gives consistent results and can handle more than one (order-restricted) hypothesis at once.

Elaborate Example to Illustrate the Confirmatory Model Selection Techniques

The final example will be used to further illustrate the (preferred) confirmatory model selection techniques, that is, the ORIC and (confirmatory) BMS.

As mentioned in the introduction, the evaluation of the order-restricted hypothesis (like H_{C1}) requires competing hypotheses. Until now, the traditional null ($H_0 : \mu_1 = \dots = \mu_k$) and alternative ($H_A : \mu_1, \dots, \mu_k$) hypotheses are used. However, it is also possible to specify a set of hypotheses without the classical null and even include another order-restricted hypothesis. We suggest that the set of hypotheses should consist of one or two order-restricted hypotheses and the unconstrained hypotheses (as a safeguard). Namely, H_0 is often not of real interest, and sometimes there is more than one theories/expectations, as in the example described next.

Gupta et al. (2008) studied the impact of implicit and explicit activation of gender stereotypes on men’s and women’s intentions to pursue an entrepreneurship, a traditionally masculine career. The variable of interest is entrepreneurial intentions (see Table 13), which is measured by the average of four items on a 5-point scale. Here, we look at 3 conditions (control, explicit masculine stereotype, and implicit masculine stereotype) for both men and women. In the control group, “participants read an article about entrepreneurship education that made no mention of gender or gender differences in entrepreneurship” (Gupta et al., 2008, p. 1055). In the other

Table 13: *Number of Observations (n_i), Sample Means ($\bar{y}_i$), and Sample Standard Deviations (sd_i) of Entrepreneurial Intentions for Group i*

i	n_i	$\bar{y}_i$	sd_i
1	38	3.44	1.01
2	46	2.94	1.07
3	36	3.48	1.08
4	37	2.66	1.09
5	33	2.93	1.05
6	39	2.43	1.00

two groups, the participants read an article in which three masculine characteristics (i.e., aggressive, risk taking, and autonomous) are mentioned. In the implicit condition, the article simply described the three characteristics, whereas in the explicit condition, there was more emphasis on the characteristics. For example, the participants in the explicit masculine stereotype group were told that it pays to have masculine characteristics. Thus, we look at six groups (see Table 14).

According to Gupta et al. (2008), previous studies show that people tend to behave in a way similar to that predicted by the stereotype they are made aware of (Theory 1). However, Gupta et al. also stated that other recent evidence suggests that under certain circumstances, people may not assimilate with the stereotype, but respond in a way opposite to that predicted by the stereotype. The response depends on whether the stereotype is activated implicitly or explicitly: Implicit stereotype activation leads to behavior consistent with the stereotype, whereas explicit activation leads to behavior opposite to the stereotype (Theory 2).

Based on the above theories, two (main) expectations are distinguished.

Table 14: *The Six Groups in Entrepreneurial Intentions*

Gender	Masculine stereotype		
	Control	Explicit	Implicit
Male	1	2	3
Female	4	5	6

The first one is based on Theory 1 and the second one is based on the expectations of Gupta et al. (2008), which are based on Theory 2. Based on these two expectations, two hypotheses are formulated, namely H_1 and H_2 , respectively. In H_1 , it is hypothesized that men who are made aware of an explicit prevalent masculine stereotype (Group 2) have the strongest entrepreneurial intentions with respect to the other same-sex groups (Groups 1 and 3). It is also hypothesized that women who are made aware of an explicit prevalent masculine stereotype (Group 5) have the weakest entrepreneurial intentions with respect to the other same-sex groups (Groups 4 and 6, respectively). So, it is hypothesized that $\mu_2 > \mu_1$, $\mu_2 > \mu_3$, $\mu_5 < \mu_4$, and $\mu_5 < \mu_6$, that is, $\mu_1 < \mu_2 > \mu_3$ and $\mu_4 > \mu_5 < \mu_6$. Furthermore, it is expected that, in the control condition, men (Group 1) have stronger entrepreneurial intentions than women (Group 4), that is, $\mu_1 > \mu_4$ (Gupta et al., 2008). Note that the expectations $\mu_2 > \mu_1$, $\mu_1 > \mu_4$, and $\mu_4 > \mu_5$ imply that $\mu_2 > \mu_5$. Therefore, it is also (indirectly) expected that in the explicit condition, men (Group 2) have stronger entrepreneurial intentions than women (Group 5), that is, $\mu_2 > \mu_5$. In the implicit condition (Group 3 and 6), however, it is unclear beforehand whose entrepreneurial intentions will be stronger. In H_2 , it is expected that “gender and stereotype activation will interact such that men will report stronger entrepreneurial intentions when presented with an implicit” (Group 3) versus an explicit (Group 2) “masculine stereotype whereas women will report stronger entrepreneurial intentions when presented with an explicit” (Group 5) versus an implicit (Group 6) masculine stereotype (Gupta et al., 2008, p. 1055), that is, $\mu_3 > \mu_2$ and $\mu_5 > \mu_6$, respectively. Furthermore, it is expected that, in the control condition, men (Group 1) will report stronger entrepreneurial intentions than women (Group 4), that is, $\mu_1 > \mu_4$. Thus, the set of hypotheses is:

$$H_1 : \begin{array}{ccccc} \mu_1 & < & \mu_2 & > & \mu_3 \\ \vee & & \vee & & , \\ \mu_4 & > & \mu_5 & < & \mu_6 \end{array}$$

$$H_2 : \begin{array}{ccccc} \mu_1 & , & \mu_2 & < & \mu_3 \\ \vee & & , & & , \\ \mu_4 & , & \mu_5 & > & \mu_6 \end{array}$$

$$H_A : \mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6,$$

where the comma denotes that there is no restriction between the correspond-

Table 15: *ORIC of the Three Specified Hypotheses (H_m) for Entrepreneurial Intentions*

m	H_m	ORIC
1	H_1	686.31
2	H_2	677.14
A	H_A	680.13

Note. ORIC = order-restricted information criterion.
Bolding indicates the lowest value.

ing means. For example, in H_1 there is no restriction between the means of Group 3 and 6. Note that the unconstrained hypothesis (H_A) is included to protect against choosing the best of two weak order-restricted hypotheses. As mentioned before, when evaluating hypotheses H_1 and/or H_2 , the pairwise relations will be evaluated simultaneously.

For the three hypotheses, the ORIC values are given in Table 15 and the PMPs in Table 16. These tables show that both ORIC and BMS prefer H_2 , the hypothesis based on the expectations of Gupta et al. (2008). Furthermore, Table 16 shows that, if a hypothesis contains only inequality constraints (i.e., “<” and/or “>”), BMS is not sensitive to the choice of the prior.

To conclude, a direct evaluation of two competing hypotheses renders strong support for H_2 . Since H_A was also included in the evaluation, H_2 is more than only the best of two weak hypotheses. This example further illustrates how ORIC and BMS can straightforwardly be used to evaluate the expectations of a researcher. All a researcher has to do is to specify expectations, collect data, and compute the ORIC or do BMS.

Discussion

Violations of the Model Assumptions

As mentioned in the introduction, all techniques are based on the ANOVA model, which has three assumptions. We first assumed that these assumptions were met. But, what if there are some violations of these assumptions? It is known that the ANOVA F test is robust with respect to nonnormality (Stevens, 1999, pp. 74–80; Toothaker, 1993, pp. 57–66). It is also robust

against violations of equal variances, as long as the ratio of the largest group size versus the smallest group size is smaller than 1.5 (Stevens, 1999, pp. 75–76). If the large variances are associated with the small group sizes, the F tests rejects the null too often; that is, the actual α is bigger than the nominal α , which is often set to .05. In contrast, when the large variances are associated with the large group sizes, the actual α is smaller than the nominal α . The ANOVA model is, however, affected by the violation of independent observations. Even a small violation results in a substantial effect on the actual significance level and power of the F test. The same is expected for the other techniques, although currently no results are available on this topic. In case of dependent observations, one could use a multilevel model (Hox, 2002). The robustness of the techniques with respect to nonnormality and unequal variances are described next and are summarized in Table 17.

In general, most multiple comparison procedures seem to be robust to moderate violations of the normality assumption (Toothaker, 1993, pp. 64–66). Miller (1986) states that the studentized range test is a bit more sensitive to nonnormality than the F test. However, as long as the number of observations per group are large enough, the studentized range statistic q should be approximately correct (according to the central limit theorem). Among others, Martin, Toothaker, and Nixon (1989) examined multiple comparison procedures with unequal variances. Their findings are the same as the findings for the ANOVA F test. Because the SWFq and TK test are multiple comparison procedures, which are based on the studentized range statistic, we expect that the above also holds for these two tests. However, further research is needed.

Table 16: *Posterior Model Probabilities (PMP) of the Three Specified Hypotheses (H_m) for Entrepreneurial Intentions (for Three Types of Prior Vagueness (PV))*

m	H_m	<i>PMP</i>		
		$PV = 1$	$PV = 2$	$PV = 3$
1	H_1	.00	.00	.00
2	H_2	.89	.89	.89
A	H_A	.11	.11	.11

Note. Bolding indicates the highest value in each column.

Table 17: *Robustness of Techniques Against Violations of Two ANOVA Assumptions*

Violations of		
	Normality	Equal variances
ANOVA F	Robust to moderate violations.	Robust as long as $\frac{\text{largest group size}}{\text{smallest group size}} < 1.5$.
stud. range (q)	More sensitive to nonnormality than the F test.	Robust as long as $\frac{\text{largest group size}}{\text{smallest group size}} < 1.5$.
SWFq	Probably resembles q , but further research needed.	Probably resembles q , but further research needed.
TK	Probably resembles q , but further research needed.	Probably resembles q , but further research needed.
$\bar{F}$	Probably resembles F , but research needed.	Probably resembles F , but research needed.
PCIC	Seems to be quite robust; further research needed.	Negative impact on the correct identification rates.
ORIC	Might resemble PCIC, but research needed.	Might resemble PCIC, but research needed.
BMS	Might resemble PCIC, but research needed.	Might resemble PCIC, but research needed.
<i>Note.</i> ANOVA = analysis of variance; SWFq = Shaffer-Welch Fq test; TK = Tukey-Kramer test; PCIC = paired-comparison information criterion; ORIC = order-restricted information criterion; BMS = Bayesian model selection.		

One could expect that the same holds for the $\bar{F}$ test as for the F test. However, as far as we know, there is no research about the violations of the ANOVA assumptions for the $\bar{F}$ test yet.

Some study has been done with respect to the robustness of the PCIC against nonnormality and heterogeneity (Dayton, 2003), but the PCIC has not been evaluated extensively. The PCIC seems to be quite robust for nonnormality. However, heterogeneity of variance had a negative impact on the correct identification rates in a simple simulation study. Note that there is a modification of the PCIC for heterogeneity of variance (see Dayton, 1998, 2003). Further research is needed for the PCIC.

One could expect that the same holds for the ORIC as for the PCIC. However, as far as we know, this is not been studied yet. The same holds for BMS. So, for the ORIC and BMS, research with respect to the violations of the assumptions is needed.

Check on Normality and Equal Variances in the Two Examples

The check on the assumption of normality and equal variances can be done by looking at the Shapiro-Wilk test and the Levene's test, respectively. For all data examples, the assumptions of normality (per group) and of equal variances are not violated, since the p values of the corresponding tests are larger than 0.05 (see Table 18). Thus, the results in this article are not influenced by violations of the assumptions.

Final Conclusions and Recommendations

Hypothesis testing techniques can give inconsistent results. Some exploratory methods cannot handle unequal group sizes, and the confirmatory approach cannot handle multiple order-restricted hypotheses. Therefore, model selection is preferred over hypothesis testing.

Confirmatory model selection techniques have a higher probability of choosing the best hypothesis than their exploratory counterparts. Furthermore, in confirmatory model selection, a direct evaluation of competing hypotheses, that is, the expectations of a researcher, is possible. However, confirmatory model selection techniques can only be used when a researcher is able to formalize his expectations in order-restricted hypotheses. To ensure that confirmatory model selection does not result in preferring a weak order-restricted hypothesis, the unconstrained hypothesis $H_A : \mu_1, \dots, \mu_k$

Table 18: *p* Values of the Tests Used for the Check on Normality (i.e., Shapiro-Wilk) and Equal Variances (i.e., Levene's) of the Two Data Examples (Defective Education and Entrepreneurial Intentions)

		Defective education		Entrepreneurial intentions
		$n_3 = 31$	$n_3 = 20$	
Shapiro-Wilk test	Group 1	.255	.255	.210
	Group 2	.446	.446	.903
	Group 3	.423	.994	.729
	Group 4			.554
	Group 5			.989
	Group 6			.839
Levene's		.135	.186	.991

must be included in the set of hypotheses. Often the classical null hypothesis is not of interest. When that is the case, the classical null hypothesis should be left out of the set of hypotheses.

Because BMS needs a specification of δ and PV (see the section of BMS) and it is a time-consuming technique, we prefer the PCIC in exploration and the ORIC in confirmation.

References

- Anraku, K. (1999). An information criterion for parameters under a simple order restriction. *Biometrika*, 86, 141–152.
- Chen, M., & Sungduk, K. (2008). Bayesian evaluation of informative hypotheses. In H. Hoijtink, I. Klugkist, & P. Boelen (Eds.), (chap. The Bayes factor versus other model selection criteria for the selection of constrained models, pp. 155–180). New York, NY: Springer.
- Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112, 155–159.
- Dayton, C. M. (1998). Information criteria for paired-comparison problem. *American Statistician*, 52, 144–151.
- Dayton, C. M. (2001). Subset: Best subsets using information criteria. *Journal of Statistical Software*, 6, 1–10.

- Dayton, C. M. (2003). Information criteria for pairwise comparisons. *Psychological Methods*, 8, 61–71.
- Gupta, V. K., Turban, D. B., & Bhawe, N. M. (2008). The effect of gender stereotype activation on entrepreneurial intentions. *Journal of Applied Psychology*, 93, 1053–1061.
- Hasel, L. E., & Kassin, S. M. (2009). On the presumption of evidentiary independence: Can confessions corrupt eyewitness identifications? *Psychological Science*, 20, 122–126.
- Hoijtink, H., Huntjens, R., Reijntjes, A., Kuiper, R., & Boelen, P. A. (2008). Bayesian evaluation of informative hypotheses. In H. Hoijtink, I. Klugkist, & P. Boelen (Eds.), (chap. An evaluation of Bayesian inequality constrained analysis of variance, pp. 85–108). New York, NY: Springer.
- Hox, J. (2002). *Multilevel analysis. techniques and applications*. Mahwah, NJ: Erlbaum.
- Johnson, V. E. (2005). Bayes factors based on test statistics. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67, 689–701.
- Klugkist, I. (2008). Bayesian evaluation of informative hypotheses. In H. Hoijtink, I. Klugkist, & P. Boelen (Eds.), (chap. Encompassing prior based model selection for inequality constrained analysis of variance, pp. 53–83). New York, NY: Springer.
- Klugkist, I., Kato, B., & Hoijtink, H. (2005). Bayesian model selection using encompassing priors. *Statistica Neerlandica*, 59, 57–69.
- Klugkist, I., Laudy, O., & Hoijtink, H. (2005). Inequality constrained analysis of variance: A bayesian approach. *Psychological methods*, 10, 477–493.
- Kramer, C. Y. (1956). Extension of multiple range tests to group means with unequal numbers of replications. *Biometrics*, 12, 307–310.
- Kramer, C. Y. (1957). Extension of multiple range tests to group correlated adjusted means. *Biometrics*, 13, 13–18.
- Lievens, F., & Sanchez, J. I. (2007). Can training improve the quality of inferences made by raters in competence modeling? a quasi-experiment. *Journal of Applied Psychology*, 92, 812–819.
- Lucas, J. W. (2003). Status processes and the institutionalization of women as leaders. *American Sociological Review*, 68, 464–480.
- Martin, S. A., Toothaker, L. E., & Nixon, S. J. (1989). A monte carlo comparison of multiple comparison procedures under optimal and nonoptimal conditions. In *a paper presented at the annual meeting of the*

southwestern psychological association, houston, tx.

- Miller, R., Jr. (1986). *Beyond anova: Basics of applied statistics*. New York, NY: Wiley.
- Neath, A. A., & Cavanaugh, J. E. (2006). A bayesian approach to the multiple comparisons problem. *Journal of Data Science*, 4, 131–146.
- Palmer, E. J., & Gough, K. (2007). Childhood experiences of parenting and causal attributions for criminal behavior among young offenders and non-offenders. *Journal of Applied Social Psychology*, 37, 790–806.
- Ramsey, P. H. (2002). Comparison of closed testing procedures for pairwise testing of means. *Psychological Methods*, 7, 504–523.
- Rossell, D., Baladandayuthapani, V., & Johnson, V. E. (2008). Bayesian evaluation of informative hypotheses. In H. Hoijtink, I. Klugkist, & P. Boelen (Eds.), (chap. Bayes factors based on test statistics under order restrictions, pp. 111–129). New York, NY: Springer.
- Shaffer, J. P. (1979). Comparison of means: An f test followed by a modified multiple range procedure. *Journal of Educational Statistics*, 4, 14–23.
- Silvapulle, M. J., & Sen, P. K. (2005). *Constrained statistical inference*. Hoboken, NJ: Wiley.
- Stevens, J. (1999). *Intermediate statistics: A modern approach*. Mahwah, NJ: Erlbaum.
- Toothaker, L. E. (1993). *Multiple comparison procedures*. Newbury Park, CA: Sage.
- Tukey, J. W. (1953). *The problem of multiple comparisons*. Unpublished manuscript, Princeton University.
- Welsch, R. E. (1977). Stepwise multiple comparison procedures. *Journal of the American Statistical Association*, 72, 566–575.
- Wiener, R. L., Holtje, M., Winter, R. J., Cantone, J. A., & Gross, K. (2007). Consumer credit card use: The roles of creditor disclosure and anticipated emotion. *Journal of Experimental Psychology: Applied*, 13, 32–46.

Appendix: Technical Notes

The Prior

The prior for the unconstrained hypothesis (i.e., the traditional alternative hypotheses) $H_A : \mu_1, \dots, \mu_k$ is defined as

$$p_A(\mu_1, \dots, \mu_k, \sigma^2) = p(\mu_1) \times \dots \times p(\mu_k) \times p(\sigma^2),$$

where $p(\mu_i) = p(\mu)$ for all i ($i = 1, \dots, k$). $p(\mu)$ is a data-based normal distribution with hyperparameters β_0 and τ_0^2 and $p(\sigma^2)$ an scaled inverse chi-squared distribution with hyperparameters ν_0 and κ_0^2 .

The prior $p_m(\cdot)$ for hypothesis H_m is (up to a normalizing constant) equal to this prior on the admissible space of H_m (and to zero outside this space). For example, in exploration, the admissible space of $H_E : \mu_1 = \mu_2$ are those combinations of means for which it holds that $|\mu_1 - \mu_2| = 0$. Note that with $|\mu_1 - \mu_2| < \delta$ an “about equality constraint” is inspected, if δ is set to a positive number. For example, in confirmation, the admissible space of $H_C : \mu_1 > \mu_2$ are those combinations of means for which it holds that $\mu_1 > \mu_2$. Here we see that, in BMS, the (order) restrictions are taken into account via the admissible space of the prior for hypothesis H_m . The admissible space covers a certain proportion of the total space, say $b\%$ (for $b \in [0, 100]$); then the prior density of H_m equals $\frac{100}{b}$ times the prior density of $H_A : \mu_1, \mu_2$ on this admissible space and equals zero outside this space. For H_C , $b = 50$, because it covers half of the total space.

Data-based Hyperparameters

For each μ_i ($i = 1, \dots, k$), a credibility interval is computed:

$$\bar{y}_i \pm PV \times \hat{\sigma}.$$

Three vague priors are used, namely the priors where PV is set to 1, 2, and 3. Note that if $PV = 3, 2$, and 1, the 99.7%, 95%, and 68% credibility intervals are computed, respectively. The credibility interval for μ_i has a lower bound LB_i and an upper bound UB_i . Let the lowest lower bound be LB_{min} (i.e., $LB_{min} = \min\{LB_1, \dots, LB_k\}$) and the highest upper bound UB_{max} . Then, the data-based hyperparameters β_0 and τ_0^2 are calculated by:

$$\beta_0 = \frac{LB_{min} + UB_{max}}{2}, \text{ and}$$

$$\tau_0^2 = \left[\frac{UB_{max} - LB_{min}}{2} \right]^2,$$

respectively. The hyperparameter ν_0 is set to 1 and κ_0^2 to the estimate of the error variance.

The Marginal Likelihood (ML)

The marginal likelihood of hypothesis H_m is defined by

$$\text{ML}(y|H_m) =$$

$$\int \dots \int \int L_m(\mu_1, \dots, \mu_k, \sigma^2|y) p_m(\mu_1, \dots, \mu_k, \sigma^2) d\mu_1 \dots d\mu_k d\sigma^2.$$

The marginal likelihood can be rewritten as

$$\text{ML}(y|H_m) = L_m(\mu_1, \dots, \mu_k, \sigma^2|y) \frac{p_m(\mu_1, \dots, \mu_k, \sigma^2)}{\text{post}_m(\mu_1, \dots, \mu_k, \sigma^2|y)},$$

where $\text{post}_m(\mu_1, \dots, \mu_k, \sigma^2|y)$ is the posterior density, for which it holds that

$$\text{post}_m(\mu_1, \dots, \mu_k, \sigma^2|y) \propto L_m(\mu_1, \dots, \mu_k, \sigma^2|y) p_m(\mu_1, \dots, \mu_k, \sigma^2).$$

Therefore,

$$-2 \log \text{ML}(y|H_m) =$$

$$-2 \log L_m(\mu_1, \dots, \mu_k, \sigma^2|y) + 2 \left[-\log \left(\frac{p_m(\mu_1, \dots, \mu_k, \sigma^2)}{\text{post}_m(\mu_1, \dots, \mu_k, \sigma^2|y)} \right) \right].$$

Posterior Model Probability (PMP)

Assuming a priori that all the hypotheses have equal probabilities of being the best (i.e., $\text{app}_m = \text{app}_{m'}$ for $m, m' = 1, \dots, M$, with M the total number of hypotheses), the posterior model probability of hypothesis $H_{m'}$ is defined as

$$\text{PMP}_{m'} = \frac{\text{app}_{m'} \text{ML}(y|H_{m'})}{\sum_{m=1}^M \text{app}_m \text{ML}(y|H_m)} = \frac{\text{ML}(y|H_{m'})}{\sum_{m=1}^M \text{ML}(y|H_m)}.$$

The $\bar{F}$ Statistic

The $\bar{F}$ statistic for testing H_{null} against H_{alt} is calculated by

$$\bar{F} = \frac{RSS(H_{null}) - RSS(H_{alt})}{S^2},$$

where S^2 is the mean square error:

$$S^2 = (n_1 + \dots + n_k - k)^{-1} \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

and $RSS(H)$ is the residual sum of squares with respect to hypothesis H ,

in an ordered alternative (i.e., $H_{null} = H_0$ and $H_{alt} = H_m$)	in an ordered null (i.e., $H_{null} = H_m$ and $H_{alt} = H_A$)
$RSS(H_{null}) = \sum_i \sum_j (y_{ij} - \bar{y})^2,$	$RSS(H_{null}) = \sum_i \sum_j (y_{ij} - \hat{\mu}_{mi})^2$
$RSS(H_{alt}) = \sum_i \sum_j (y_{ij} - \hat{\mu}_{mi})^2,$	$RSS(H_{alt}) = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2,$

with $\bar{y}$ as the overall mean, $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$ as the restricted mean values, that is, the values of $\mu_1, \dots, \mu_k$ that minimize $\sum_i \sum_j (y_{ij} - \mu_i)^2$, subject to $H_m : \mu_i - \mu_{i'} \geq 0$ for some $i, i' = 1, \dots, k$, and $\bar{y}_i$ as the i th group mean. More details about the restricted means can be found in the next section.

Note that the classical F test is based on $RSS(H_{null}) = \sum_i \sum_j (y_{ij} - \bar{y})^2$ and $RSS(H_{alt}) = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2$.

The Restricted Means

As mentioned, the restricted means (i.e., $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$) are in accordance with the hypotheses of interest, H_m . This was also the case with the restricted means used in the PCIC, only now H_m can contain order restrictions.

For $H_1 : \mu_1 > \dots > \mu_k$, it holds that $\hat{\mu}_{11} \geq \dots \geq \hat{\mu}_{1k}$ and their values depends on the (weighted) sample means. An example is given in Table 19. To elaborate on this, numerical examples for $k = 3$ and equal group sizes are given in Table 20.

When the restrictions are simple order restrictions (i.e., the restrictions can be written as $\mu_1 > \dots > \mu_{k'}$ for $k' \leq k$), the restricted means can be calculated by the pool adjacent violators algorithm (PAVA). For more details see Silvapulle and Sen (2005, pp. 47–50). Another method, which can be applied to all types of order restrictions, is quadratic programming. For more details see Silvapulle and Sen (2005, pp. 36–37).

Table 19: *The Restricted Mean Values for Different Sample Mean Values When $H_1 : \mu_1 > \dots > \mu_k$*

Ordering of the sample means	Corresponding restricted means				
	$\hat{\mu}_{11}$	$\dots$	$\hat{\mu}_{1k-2}$	$\hat{\mu}_{1k-1}$	$\hat{\mu}_{1k}$
$\bar{y}_1 \geq \dots \geq \bar{y}_k$	$\bar{y}_1$	$\dots$	$\bar{y}_{k-2}$	$\bar{y}_{k-1}$	$\bar{y}_k$
$\bar{y}_1 \geq \dots \geq \bar{y}_{k-2}$ and $\bar{y}_{k-1} \leq \bar{y}_k$	$\bar{y}_1$	$\dots$	$\bar{y}_{k-2}$	$\frac{n_{k-1}\bar{y}_{k-1} + n_k\bar{y}_k}{n_{k-1} + n_k}$	$\frac{n_{k-1}\bar{y}_{k-1} + n_k\bar{y}_k}{n_{k-1} + n_k}$

Table 20: *Numerical Examples of the Restricted Mean Values for $k = 3$ and for Equal Group Sizes When $H_1 : \mu_1 > \dots > \mu_k$*

Sample means			Corresponding restricted means		
$\bar{y}_1$	$\bar{y}_2$	$\bar{y}_3$	$\hat{\mu}_{11}$	$\hat{\mu}_{12}$	$\hat{\mu}_{13}$
2	1	2	2	1.5	1.5
1	2	1.4	1.5	1.5	1.4
1	2	2	$1\frac{2}{3}$	$1\frac{2}{3}$	$1\frac{2}{3}$

Calculation of the p Value of the $\bar{F}$ Statistic

1. Generate independent observations z_{ij} ($i = 1, \dots, k$ and $j = 1, \dots, n_i$) from the standard normal distribution $\mathcal{N}(0, 1)$.
2. Compute the value of the $\bar{F}$ statistic for the generated data.
3. Repeat the previous two steps R times. R is set to $R = 100,000$, because then we get stable estimates for the p values, that is, when we calculated the p value again, the difference was rarely larger than 0.001.
4. Determine the number of times the $\bar{F}$ statistic, calculated in Step 2, exceeds the sample value of the $\bar{F}$. Denoted this by C .
5. The p value is estimated by C/R .

Calculation of the Level Probabilities

1. Generate $z_1, \dots, z_k$ from $\mathcal{N}_k(0, V)$, that is, the multivariate normal distribution, with mean 0 and the covariance matrix V , where V is a diagonal matrix with the elements $1/n_1, \dots, 1/n_k$ on the diagonal.
2. Compute the restricted means $\hat{z}_{m1}, \dots, \hat{z}_{mk}$ analogously to $\hat{\mu}_{m1}, \dots, \hat{\mu}_{mk}$. So, $\hat{z}_{m1}, \dots, \hat{z}_{mk}$ are the values for which $L_m(\hat{z}_{m1}, \dots, \hat{z}_{mk}, \hat{\sigma}_m^2 | z_1, \dots, z_k)$ (see (5)) is maximized subject to hypothesis H_m .
3. Determine the number of distinct values in $\hat{z}_{m1}, \dots, \hat{z}_{mk}$, called levels. Denote this by $\tilde{l}_m$.
4. Repeat the previous steps R times. R is set to $R = 100,000$, because in that case we get stable estimates for LP_{ml} . That is, when we calculated it again, the difference was most of the time less than 0.02.
5. Estimate the level probability LP_{ml} by the proportion of times $\tilde{l}_m$ is equal to l .

Because the restricted means $\hat{z}_{m1}, \dots, \hat{z}_{mk}$ are in accordance with H_m , the maximum value $\tilde{l}_m$ can take on is a_m . For example, for $H_2 : \mu_1 > \mu_2 = \mu_3$, it holds that $a_2 = 2$ and $\hat{z}_{21} \geq \hat{z}_{22} = \hat{z}_{23}$. When $\hat{z}_{21} = \hat{z}_{22} = \hat{z}_{23}$, there is one level, and when $\hat{z}_{21} > \hat{z}_{22} = \hat{z}_{23}$, there are two levels. Therefore, for $l = a_m + 1, \dots, k$, $LP_{ml} = 0$.